

Phase 1

Fundamentals — Index & Study Checklist

Goal: build every foundational HLD block from zero — no prior system design knowledge assumed.
Complete chapters in order; each later chapter assumes earlier ones.

#	Chapter	Time
1	HLD vs LLD What HLD is, how it differs from LLD and coding interviews	1 hr
2	Client-Server Model & Request Lifecycle The request/response model underlying every system	1.5 hr
3	DNS & How the Internet Really Works Domain resolution, TTLs, DNS-based routing	1 hr
4	Networking Essentials TCP vs UDP, HTTP/HTTPS, WebSockets	2 hr
5	APIs & Communication Contracts REST vs gRPC, sync vs async	1.5 hr
6	Databases 101 — Relational ACID, transactions, indexing	2 hr
7	Databases 101 — NoSQL Key-value, document, column-family, graph	2 hr
8	Caching Fundamentals Cache-aside, write-through, write-back, eviction	1.5 hr
9	Load Balancing Basics L4 vs L7, algorithms, health checks	1.5 hr
10	Proxies & CDNs Forward vs reverse proxy, CDN mechanics	1 hr
11	Object / Blob Storage Basics S3-style storage vs databases vs local disk	1 hr
12	Introduction to Scalability Vertical vs horizontal scaling, statelessness	1.5 hr
13	Non-Functional Requirements & Latency Numbers SLI/SLO/SLA, percentiles, latency intuition	1.5 hr
14	Back-of-the-Envelope Estimation Estimating QPS, storage, bandwidth	1.5 hr

#	Chapter	Time
15	Phase 1 Recap & Mini System Capstone: design a pastebin end to end	1.5 hr

Phase 1 total: ~19 hours

Study Checklist

Mark each chapter complete only once you can do all three of the following without notes:

- Explain the core concept out loud in under 2 minutes, using the backend-engineer analogy as a bridge.
- Answer both "Common Interview Questions" from memory with a full model-approach structure.
- Complete the practice exercise(s) unaided.

Done?	Chapter
☐ Chapter 1	HLD vs LLD
☐ Chapter 2	Client-Server Model & Request Lifecycle
☐ Chapter 3	DNS & How the Internet Really Works
☐ Chapter 4	Networking Essentials
☐ Chapter 5	APIs & Communication Contracts
☐ Chapter 6	Databases 101 — Relational
☐ Chapter 7	Databases 101 — NoSQL
☐ Chapter 8	Caching Fundamentals
☐ Chapter 9	Load Balancing Basics
☐ Chapter 10	Proxies & CDNs
☐ Chapter 11	Object / Blob Storage Basics
☐ Chapter 12	Introduction to Scalability
☐ Chapter 13	Non-Functional Requirements & Latency Numbers
☐ Chapter 14	Back-of-the-Envelope Estimation
☐ Chapter 15	Phase 1 Recap & Mini System

How Phase 1 Chapters Connect

Phase 1 is deliberately ordered so each chapter's mental model supports the next:

- 1 Ch 1 sets the frame (HLD vs LLD) so every later chapter is understood at the right altitude.
- 2 Ch 2–5 build the communication layer: how a request travels, over what protocols, in what contract shape.
- 3 Ch 6–7 build the storage layer: where data lives and which guarantees it has.
- 4 Ch 8–11 build the performance/delivery layer: caching, load balancing, proxies/CDN, object storage.
- 5 Ch 12–14 build the scale-reasoning layer: how to grow the system and how to quantify "how much."
- 6 Ch 15 forces you to combine all of the above into one working design — the real test of Phase 1.

Next step

Once every chapter is checked off and Chapter 15's capstone is solved unaided, you're ready to approve Phase 2 (Intermediate) — scalability patterns, consistency models, sharding, and messaging.